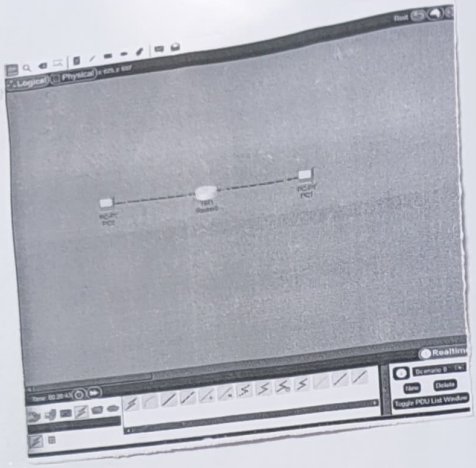
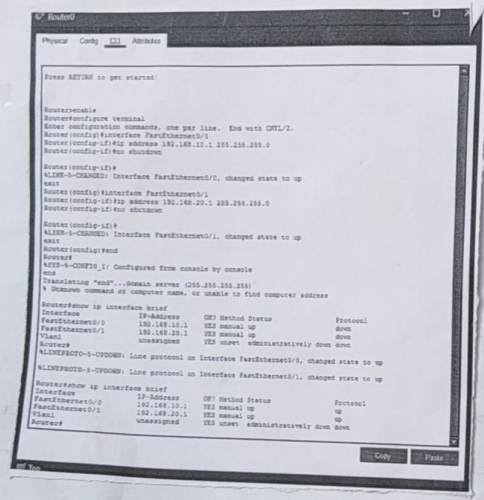




EX.NO 10 a INTERNETWORKING WITH ROUTERS IN CISCO PACKET TRACER.

AIM:

To design and configure a simple internetwork using one router and 2 PCs in cisco packet tracer and verify connectivity b/w PCs.

NETWORK CONFIGURATION:

Device	Interface	IP Address
Router 1	FastEthernet 0/0	192.168.10.1
Router 2	FastEthernet 0/1	192.168.20.1
PC0	FastEthernet 0	192.168.10.2
PC1	FastEthernet 0	192.168.20.2

PROCEDURE:

Step 1: Configure Router 1

Open Router 1 → CLI and enter:

Router > enable

Router # configure terminal

Router (config) # interface FastEthernet 0/0

Router (config-if) # ip address 192.168.10.1 255.25.255.0

Router (config-if) # no shutdown

Router (config-if) # interface FastEthernet 0/1

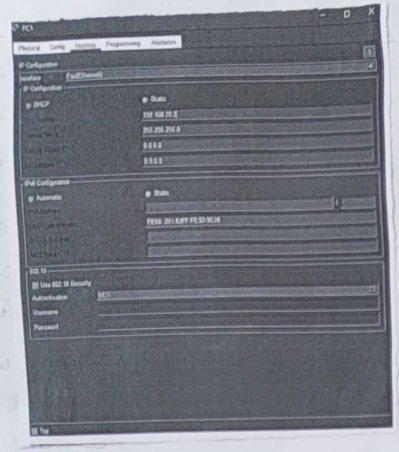
Router (config-if) # ip address 192.168.20.1 255.255.255.0

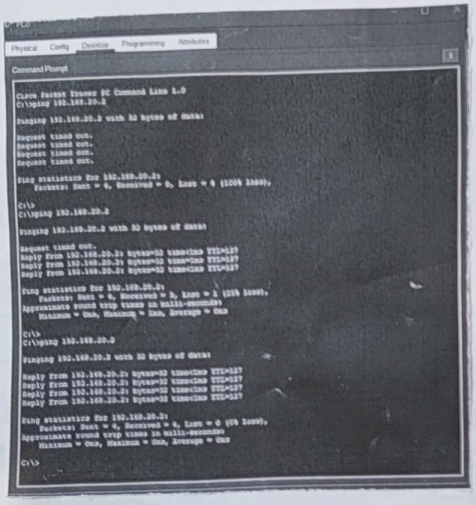
Router (config-if) # no shutdown

Router (config-if) # exit

Router (config) # exit

Router #





Step 2: Configure PC0

Go to - PC0 → Desktop → IP configuration

IP Address 192.168.10.2

Subnet mask : 255.255.255.0

Default Gateway: 192.168.10.1

Step 3: Configure PC1

Go to PC1 → Desktop → IP Configuration

IP Address : 192.168.20.2

Subnet mask : 255.255.255.0

Default Gateway: 192.168.20.1

Step 4: Connect devices

Use Copper Straight - through cables

PC0 FastEthernet 0

↓ Copper Straight Through

Router 1 FastEthernet 0/0

Router 1 FastEthernet 0/1

↓ Copper Straight Through

PC 1 FastEthernet 0

Step 5: Verify connectivity

Ping 192.168.20.2

A successful reply indicates that PC0 can communicate with PC1 through Router 1.

Also use Packet tracer:

1. Select add simple PDU envelope icon
2. Click PC0 as source
3. Click PC1 as destination
4. Check simulation window
5. A successful PDU indicates connectivity

RESULT:

A single internetwork consisting of 2 PCs and one router was successfully designed and configured in Cisco Packet Tracer